

COMMUNITY ENGAGEMENT PROJECT REPORT ON

IoT-Enabled Industrial Security System Using ESP32-S3 CAM with Real-Time Monitoring and 360° Surveillance

Project report submitted in partial fulfillment of the requirement for the degree of
Bachelor of Technology (Artificial Intelligence & Machine Learning)

By

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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE
LEARNING**

**SANJIVANI UNIVERSITY, KOPARGAON, MAHARASHTRA
(AY: 2025-26)**

DECLARATION

I hereby solemnly declare that the project report entitled “**IoT-Enabled Industrial Security System Using ESP32-S3 CAM with Real-Time Monitoring and 360° Surveillance**”, submitted to the Department of Artificial Intelligence and Machine Learning (AIML), Sanjivani University, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering (AIML)**, is a genuine and original record of the work carried out by me.

This project has been completed under the valuable guidance, continuous support, and supervision of **Dr. Monika Arya**, whose expertise, encouragement, and constructive feedback have been instrumental throughout the development of this work. I express my sincere gratitude for her constant motivation and insightful suggestions, which have significantly contributed to the successful completion of this project.

I further declare that this work has not been submitted, either in part or in full, to any other university, institution, or organization for the award of any degree, diploma, associateship, fellowship, or any other similar title. The content presented in this report is solely based on my own efforts, research, and implementation, except where due acknowledgment has been made in the text.

I also confirm that all sources of information, data, and references used in this project have been properly cited and acknowledged as per academic standards. Any assistance received during the course of this work has been duly recognized.

I take full responsibility for the authenticity and accuracy of the information presented in this report.

Dated

Signature of the Candidate

Certificate

This is to certify that the project report entitled “**IoT-Enabled Industrial Security System Using ESP32-S3 CAM with Real-Time Monitoring and 360° Surveillance**” submitted by **Soham Prabhakar Kadu** to the Department of Artificial Intelligence and Machine Learning (AIML), Sanjivani University, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering (AIML)**, is a bona fide record of work carried out by him.

The project work has been completed under my supervision and guidance during the academic year, and to the best of my knowledge and belief, the work presented in this report is original and has not been submitted, either in part or in full, to any other university, institution, or organization for the award of any degree, diploma, associateship, fellowship, or any other similar title.

The candidate has demonstrated sincere effort, dedication, and a good understanding of the subject during the course of this project work.

Signature of

Guide: _____

Name: Dr. Monika Arya

Department: Computer Science &
Engineering (Artificial Intelligence
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Signature of Head of Department:

Name: Dr. Devyani Jadhav

Department: Computer Science & Engineering
(Artificial Intelligence & Machine Learning)

Institution: Sanjivani University, Kopergaon

Abstract

Industrial security is a major concern these days. Theft, unauthorized entry, and environmental damage cause huge losses every year. Traditional CCTV systems are expensive and don't give real-time sensor data. They also don't allow remote control.

We built a low-cost, IoT-based security system using the XIAO ESP32-S3 Cam. It has a live video stream over Wi-Fi, a PIR motion sensor (can be replaced with an ultrasonic sensor for distance-based detection), a DHT11 for temperature/humidity, an LDR for light, and a rain sensor. The camera is mounted on a 360° continuous servo motor that slowly pans to cover the whole area. When motion is detected, a buzzer and a red LED turn on for 5 seconds. All data and controls are on a web dashboard – you can watch live video, see sensor values, pan the camera manually, adjust pan speed, and turn LEDs/buzzer on/off from any browser.

The total cost is less than ₹3000. The system works well indoors and outdoors. It can be used in factories, warehouses, farms, or even smart homes. The response time for motion alert is under 0.5 seconds, and video latency is about 200 ms. This project shows that effective industrial security can be achieved with very affordable hardware.

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CHAPTER 1: INTRODUCTION

Industrial security has become a major concern in modern environments due to increasing cases of theft, unauthorized access, and environmental hazards. Traditional surveillance systems such as CCTV cameras are widely used, but they suffer from several limitations such as fixed viewing angles, high cost, and lack of environmental monitoring.

In many industries, it is not sufficient to only monitor video footage. Parameters like temperature, humidity, light intensity, and weather conditions also play an important role in safety and operations. Existing systems do not provide an integrated solution for both surveillance and environmental monitoring.

To overcome these limitations, this project presents a **smart surveillance system using ESP32-S3 CAM**. The system integrates a camera with multiple sensors and provides real-time monitoring through a web-based dashboard. The camera is mounted on a servo motor to achieve **360-degree coverage**, eliminating blind spots.

The system is designed to be low-cost, easy to deploy, and accessible remotely via Wi-Fi. It can be used in industries, warehouses, farms, and smart homes.

CHAPTER 2: LITERATURE SURVEY

Various surveillance systems have been developed using technologies such as CCTV, Raspberry Pi, and IoT-based devices.

Traditional CCTV systems provide continuous video monitoring but lack flexibility and intelligence. They are expensive and require complex installation. Raspberry Pi-based systems offer better flexibility but consume more power and increase system cost.

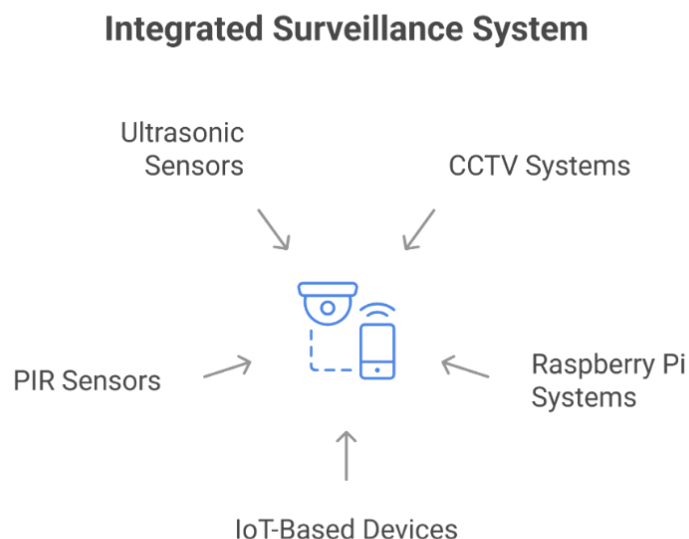
Some IoT-based projects using ESP32-CAM provide video streaming capabilities. However, most of them do not include additional sensors or automated camera movement. Similarly, sensor-based monitoring systems exist separately but are not integrated with video surveillance.

Motion detection is commonly implemented using PIR sensors, which detect infrared radiation from human bodies. Ultrasonic sensors can also be used for object detection based on distance measurement.

From the literature survey, it is observed that there is a lack of systems that combine:

- Video surveillance
- Environmental monitoring
- Remote control
- 360° camera movement

This project aims to bridge this gap by integrating all these features into a single system.



CHAPTER 3: METHODOLOGY USED

The system is developed using both hardware and software components integrated together.

System Overview

The ESP32-S3 CAM acts as the central controller. It processes sensor data, streams video, and hosts a web server.

Working Process

1. When the system is powered ON, all components are initialized.
2. The ESP32 connects to Wi-Fi.
3. The camera starts streaming video.
4. Sensors begin collecting data periodically.

Sensor Integration

- **DHT11** → Temperature & Humidity
- **LDR** → Light intensity
- **Rain Sensor** → Water detection
- **PIR Sensor** → Motion detection

Motion Detection

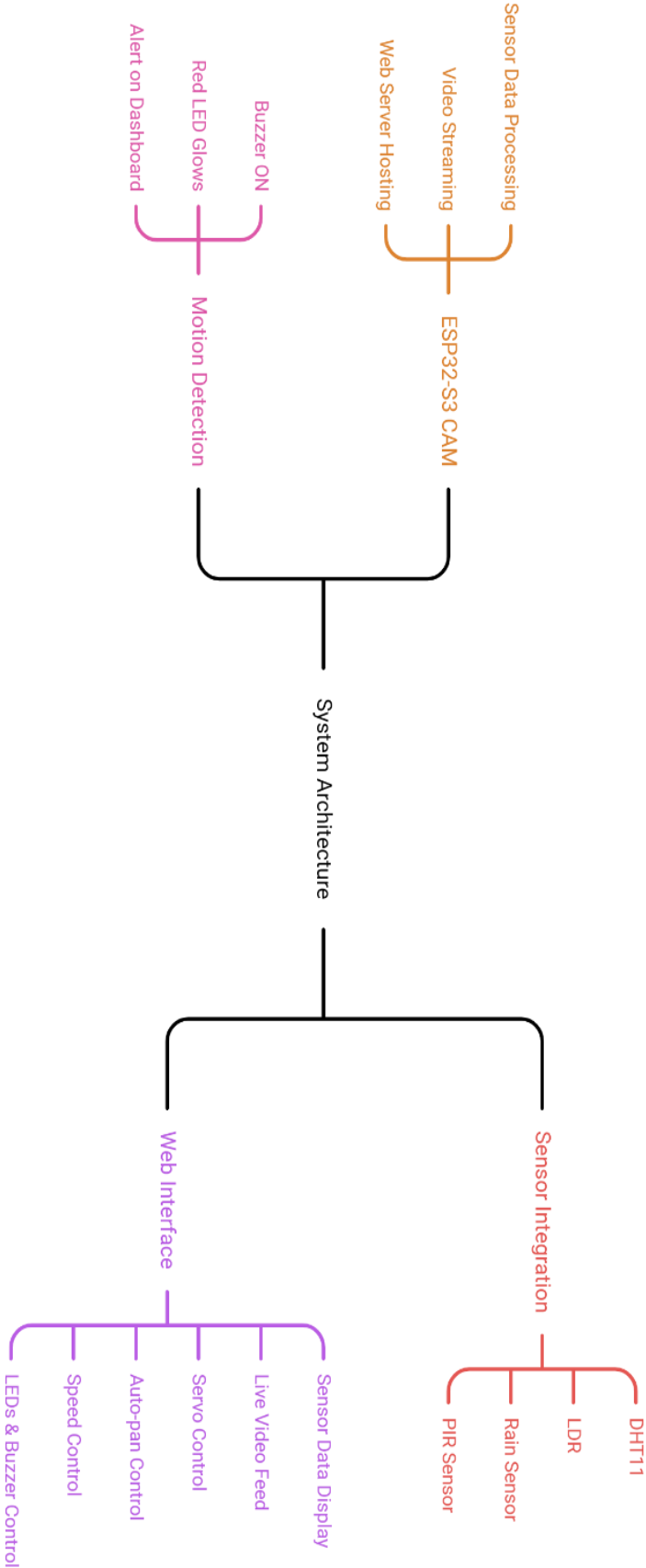
When motion is detected:

- Buzzer turns ON
- Red LED glows
- Alert shown on dashboard

Web Interface

- Displays sensor data
- Shows live video
- Allows control of:
 - Servo (Left/Right/Stop)
 - Auto-pan
 - Speed
 - LEDs & Buzzer

System Architecture and Functionality



CHAPTER 4: CODE IMPLEMENTATION

The project is implemented using Arduino IDE with ESP32 libraries.

◇ Main Functional Modules:

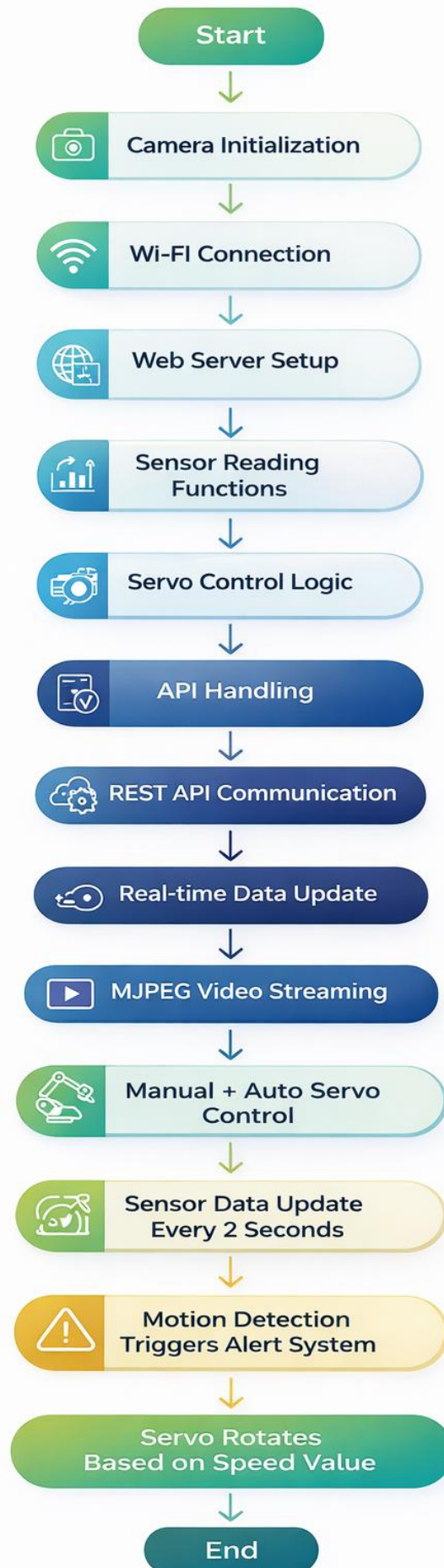
- Camera initialization
- Wi-Fi connection
- Web server setup
- Sensor reading functions
- Servo control logic
- API handling

◇ Key Features:

- REST API communication
- Real-time data update
- MJPEG video streaming
- Manual + Auto servo control

◇ Example Logic:

- Sensor data is updated every 2 seconds
- Motion detection triggers alert system
- Servo rotates based on speed value



CHAPTER 5: RESULTS / SNAPSHOTS

The system was successfully tested under different conditions.

Observations:

- Live video streaming achieved (~15 FPS)
- Sensor data updated in real time
- Motion detection response < 0.5 sec
- Servo rotation smooth and stable

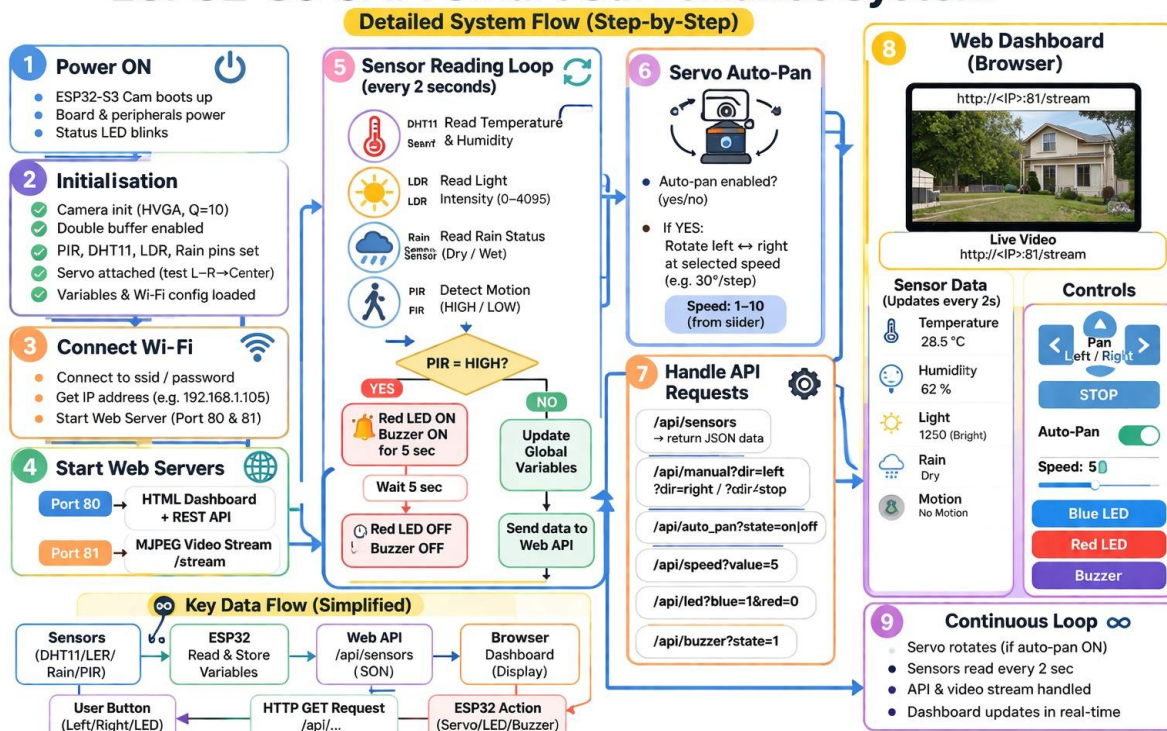
Performance:

- Latency: ~200 ms
- Range: ~30 meters
- Continuous operation stable

Functional Testing:

- Motion detection → Working
- Rain detection → Working
- Light sensing → Working
- Manual control → Working

ESP32-S3 CAM Smart Surveillance System



CHAPTER 6: CONCLUSION AND RECOMMENDATION

The project successfully demonstrates a **low-cost smart surveillance system** integrating camera, sensors, and remote control.

The system provides:

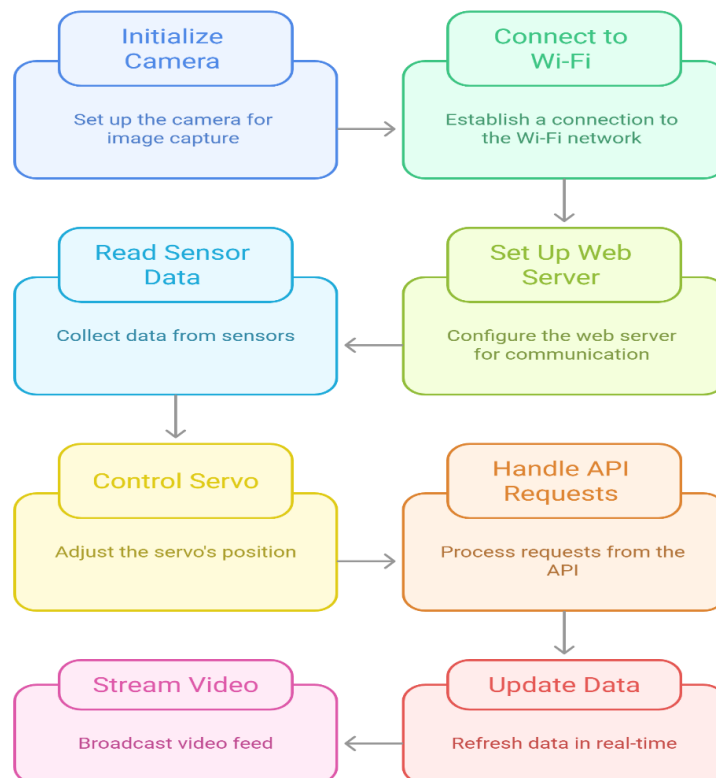
- 360° monitoring
- Real-time sensor tracking
- Instant alerts
- User-friendly interface

It is suitable for small industries, warehouses, and farms. The system is reliable and easy to maintain.

◆ Recommendations:

- Use better sensors for higher accuracy
- Improve enclosure for outdoor use
- Add backup power system

Code Implementation Sequence



FUTURE SCOPE

1. Mobile Application Integration

Develop an Android/iOS app for real-time monitoring, notifications, and remote control.

2. Cloud Storage Integration

Use platforms like Firebase, AWS IoT, or Google Cloud for storing data and enabling remote access.

3. AI-Based Object Detection

Implement AI/ML using tools like TensorFlow Lite to detect and classify objects, reducing false alarms.

4. Solar-Powered System

Integrate solar panels and batteries for energy-efficient and remote operation.

5. Multi-Camera Setup

Use multiple ESP32-CAM modules with a centralized dashboard for large-scale surveillance.

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